OpenSees Modified IMK Material Property

Disclaimer:

This notes are summarized by XINGQUAN GUAN, who is currently studying in Professor Henry Burton Research Group. (Discussions are welcome: guanxingquan@ucla.edu)

The equations are for steel beam with reduced beam section (RBS), steel beam other than RBS, and steel column.

The composite action is not considered in the modeling.

All notes are based on the papers listed in Reference section.

Reference:

[1] Elkady and Lignos. (2019). Two-dimensional OpenSees numerical models for archetype steel buildings with special moment frames. Available at: https://github.com/amaelkady/OpenSEES_Models_SMF.git

[2] Lignos and Krawinkler. (2011) Deterioration modeling of steel components in support of collapse prediction of steel moment frames under earthquake loading

[3] Ibarra and Krawinkler. (2005) Global collapse of frame structures under seismic excitations

[4] Lignos. (2008). Sidesway collapse of deteriorating structural systems under seismic excitations

[5] Lignos et al. (2019). Proposed updates to the ASCE 41 nonlinear modeling parameters for wide-flange steel columns in support performance-based seismic engineering

OpenSees IMK Command:

The command is:

uniaxialMaterial Bilin *matTag K0 as_Plus as_Neg My_Plus My_Neg Lamda_S Lamda_C Lamda_A Lamda_K c_S c_C c_A c_K theta_p_Plus theta_p_Neg theta_pc_Plus theta_pc_Neg Res_Pos Res_Neg theta_u_Plus theta_u_Neg D_Plus D_Neg*

The explanations for those parameters are available at: https://opensees.berkeley.edu/wiki/index.php/Modified_Ibarra-Medina-Krawinkler_Deterioration_Model_with_Bilinear_Hysteretic_Response_(Bilin_Material)

Modeling Parameters:

- K0

$$K_0 = (n + 1.0) \cdot 6 \cdot E \cdot I/L$$

where n is typically set as 10.0.

- as_Plus, as_Neg

$$a_{s,plus} = a_{s,neg} = \frac{a_{men}}{1 + n \cdot (1 - a_{men})}$$
$$a_{men} = (n + 1.0) \cdot \frac{M_y(M_c/M_y - 1.0)}{\theta_p} \cdot \frac{1}{K_0} == \frac{M_y(M_c/M_y - 1.0)}{\theta_p} \cdot \frac{1}{6EI/L}$$

For RBS beam and non RBS beam:

$$M_c = 1.10 \cdot M_y$$

For column:

$$M_c = a \cdot M_y$$
$$a = 12.5 \cdot (\frac{h}{t_w})^{-0.2} \cdot (\frac{L_b}{r_y})^{-0.4} \cdot (1 - \frac{P_g}{P_{ye}})^{0.4}, \quad 1.0 \leq a \leq 1.3$$

- My_Plus, My_Neg

  For RBS beam (in Lignos 2011 paper):

$$M_y = 1.06 \cdot M_{y,p} = 1.06 \cdot Z \cdot F_y$$

  For non RBS beam (in Lignos 2011 paper):

$$M_y = 1.17 \cdot M_{y,p} = 1.17 \cdot Z \cdot F_y$$

  For column (in Lignos 2019 Paper):

$$M_y = M_y^* = 1.15 \cdot Z \cdot R_y \cdot F_{yn} \cdot (1 - \frac{P_g}{2P_{ye}}) \quad \text{if } P_g/P_{ye} < 0.20$$
$$My = M_y^* = 1.15 \cdot Z \cdot Ry \cdot F_{yn} \cdot \frac{9}{8}(1 - \frac{P_g}{P_{ye}}) \quad \text{if } P_g/P_{ye} \geq 0.20$$

  For column (in Lignos OpenSees source codes)

$$M_y = M_y^* = 1.15/1.1 \cdot M_y \cdot (1 - \frac{P_g}{2P_{ye}}) \quad \text{if } P_g/P_{ye} < 0.20$$
$$My = M_y^* = 1.15/1.1 \cdot M_y \cdot \frac{9}{8}(1 - \frac{P_g}{P_{ye}}) \quad \text{if } P_g/P_{ye} \geq 0.20$$

- Lamda_S, Lamda_C, Lamda_A, Lamda_K

  For RBS beam:

$$\Lambda = \frac{E_t}{M_y} = 585 \cdot (\frac{h}{t_w})^{-1.14} \cdot (\frac{b_f}{2 \cdot t_f})^{-0.632} \cdot (\frac{L_b}{r_y})^{-0.205} \cdot (\frac{c_{unit}^2 \cdot F_y}{355})^{-0.391}$$
$$\Lambda_S = \Lambda_C = \Lambda_A = \Lambda_k$$

  If millimeters and megapascals are used:

$$c_{unit}^1 = c_{unit}^2 = 1.0$$

  If depth is in inches and F_y is in ksi:

$$c_{unit}^1 = 25.4$$
$$c_{unit}^2 = 6.895$$

  For non RBS beam:

$$\Lambda = \frac{E_t}{M_y} = 495 \cdot (\frac{h}{t_w})^{-1.34} \cdot (\frac{b_f}{2 \cdot t_f})^{-0.595} \cdot (\frac{c_{unit}^2 \cdot F_y}{355})^{-0.360}$$
$$\Lambda_S = \Lambda_C = \Lambda_A = \Lambda_k$$

For Column (in Lignos 2019 Paper):

$$\Lambda = 255000 \cdot (\frac{h}{t_w})^{-2.14} \cdot (\frac{L_b}{r_y})^{-0.53} \cdot (1 - \frac{P_g}{P_{ye}})^{4.92} \leq 3.0, \quad \text{if } P_g/P_{ye} \leq 0.35$$
$$\Lambda = 268000 \cdot (\frac{h}{t_w})^{-2.30} \cdot (\frac{L_b}{r_y})^{-1.30} \cdot (1 - P_g/P_{ye})^{1.19} \leq 3.0, \quad \text{if } P_g/P_{ye} > 0.35$$
$$\Lambda_K = \Lambda_C = \Lambda_A = 0.9 \cdot \Lambda_S$$

For column (in Lignos OpenSees source codes):

$$\Lambda_S = \Lambda_C = \Lambda_A = \Lambda_K$$

- c_S, c_A, c_C, c_K

$$c_S = c_A = c_c = c_K = 1.0$$

- theta_p_Plus, theta_p_Neg

For RBS beam:

$$\theta_p = 0.19 \cdot (\frac{h}{t_w})^{-0.314} \cdot (\frac{b_f}{2 \cdot t_f})^{-0.100} \cdot (\frac{L_b}{r_y})^{-0.185} \cdot (\frac{L}{d})^{0.113} \cdot (\frac{c_{unit}^1 \cdot d}{533})^{-0.760} \cdot (\frac{c_{unit}^2 \cdot F_y}{355})^{-0.070}$$

For non RBS beam:

$$\theta_p = 0.0865 \cdot (\frac{h}{t_w})^{-0.365} \cdot (\frac{b_f}{2 \cdot t_f})^{-0.140} \cdot (\frac{L}{d})^{0.340} \cdot (\frac{c_{unit}^1 \cdot d}{533})^{-0.721} \cdot (\frac{c_{unit}^2 \cdot F_y}{355})^{-0.230}$$

For column:

$$\theta_p = 294 \cdot (\frac{h}{t_w})^{-1.7} \cdot (\frac{L_b}{r_y})^{-0.7} \cdot (1 - \frac{P_g}{P_{ye}})^{1.6} \leq 0.20\, rad$$

After determined using three equations above, θ_p needs to be further corrected to account for elastic deformation:

$$\theta_p = \theta_p - \frac{(M_c/M_y - 1.0) \cdot M_y}{6 \cdot E \cdot I/L}$$

- theta_pc_Plus, theta_pc_Neg

For RBS beam:

$$\theta_{pc} = 9.52 \cdot (\frac{h}{t_w})^{-0.513} \cdot (\frac{b_f}{2 \cdot t_f})^{-0.863} \cdot (\frac{L_b}{r_y})^{-0.108} \cdot (\frac{c_{unit}^2 \cdot F_y}{355})^{-0.360}$$

For non RBS beam:

$$\theta_{pc} = 5.63 \cdot (\frac{h}{t_w})^{-0.565} \cdot (\frac{b_f}{2 \cdot t_f})^{-0.800} \cdot (\frac{c_{unit}^1 \cdot d}{533})^{-0.280} \cdot (\frac{c_{unit}^2 \cdot F_y}{355})^{-0.430}$$

For column:

$$\theta_{pc} = 90 \cdot (\frac{h}{t_W})^{-0.8} \cdot (\frac{L_b}{r_y})^{-0.8} \cdot (1 - \frac{P_g}{P_{ye}})^{2.5} \leq 0.30\, rad$$

After determined using the three equations above, $\theta_p c$ needs to be further corrected to account for elastic deformation:

$$\theta_{pc} = \theta_{pc} + \theta_y + \frac{(M_c/M_y - 1.0) \cdot M_y}{6 \cdot E \cdot I/L}$$
$$\theta_y = \frac{M_y}{6 \cdot E \cdot I/L}$$

- Res_Pos, Res_Neg

For RBS and non RBS beam: both of them are 0.4

For column::

$$k = M_r/M_y = (0.5 - 0.4 \cdot P_g P_{ye})$$

- theta_u_Plus, theta_u_Neg

For RBS and non RBS beam: both of them are 0.2

For column: it is 0.15

- D_Plus, D_Neg:

  All of three types of component are 1.0

Elastic beam-column element modeling parameters:

For *elastic beam-column element*, modified moment of inertia is:

$$I_{mod} = \frac{n + 1.0}{n} \cdot I_z$$

Flexural stiffness for *elastic beam-column element*:

$$K_{bc} = \frac{n + 1}{n} \cdot \frac{6 \cdot E \cdot I_z}{L}$$

In contrast, the flexural stiffness for the HINGE is (which is shown in the previous section)s:

$$K_0 = n \cdot \frac{6 \cdot E \cdot I_{mod}}{L} = (n + 1.0) \cdot \frac{6 \cdot E \cdot I_z}{L}$$